

LC 1047: Remove All Adjacent Duplicates In String

LeetCode 1047

Whatfix

Paytm

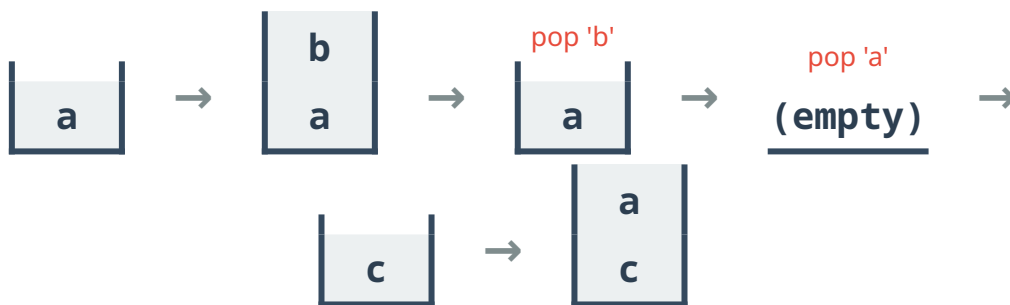
Adobe

Problem Understanding

→ choose any 2 adjacent and equal letters and remove them

eg: `abbaca` → `aaca` → `ca`

Visualizing the Stack Approach



hence for any char `c` → if `c == st.top()` pop the stack , else push it

Final Result Extraction

at last , you get a stack with answer string



C++ Implementation

```
class Solution {
public:
    string removeDuplicates(string s) {

        stack<char> st;

        for(char c : s) {
            if(!st.empty() && st.top() == c) {
                st.pop();
            }
            else {
                st.push(c);
            }
        }

        string ans = "";
        while(!st.empty()) {
            ans += st.top();
            st.pop();
        }

        reverse(ans.begin(), ans.end());
        return ans;
    }
};
```